RADIATION SAFETY PRODUCT REPORT OF ORTUR LASER MASTER PRO 3 DESKTOP ENGRAVER

PREPARED PURSUANT TO FORM FDA 3632 GUIDE & 21 CFR 1002.10 (EPRC)
CDRH Electronic Product Radiation Control • U.S. Food and Drug Administration

Manufacturer:	Dongguan Ortur Intelligent Technologies Co., Ltd.	**Date of Report:**	2026-11-02
Model Family:	OLM3-20W, OLM3-10W, OLM3-LE-5W	**Laser Class:**	**Class IV**
Wavelength(s):	450 nm	**Max Radiant Power:**	20.0 W radiant optical flux
US Agent:	EPRC US Agent Services LLC	**Submission Channel:**	RadHealthCustomerService@fda.hhs.gov

PART 1: IDENTIFICATION OF MANUFACTURER & US DESIGNATED AGENT

Manufacturer Legal Entity	Dongguan Ortur Intelligent Technologies Co., Ltd.
Operating Address	Building 1, No. 6 Tianhao Road, Dongcheng District, Dongguan, Guangdong, China, China
Regulatory Contact	Justin Zhang (Director of Engineering & Regulatory Compliance) | regulatory@ortur.cn | +86-769-82688888
US Agent for Service (21 CFR 1005.25)	EPRC US Agent Services LLC; 1209 Orange Street, Wilmington, DE 19801; compliance@eprcagent.com

PART 2: RADIATION EMISSION & CLASSIFICATION CHARACTERISTICS (21 CFR 1040.10)

Active Medium / Architecture	Semiconductor InGaN Laser Diode Array	**Operational Mode**	Continuous Wave (CW)
Emission Wavelength(s)	450 nm	**Max Output Power**	20.0 W radiant optical flux
Beam Divergence	1.2 mrad	**Beam Aperture Diameter**	5.5 mm
Conformance Standard	Laser Notice 56 / IEC 60825-1:2014 Ed. 3 & 21 CFR 1040.10	**Laser Class**	**Class IV**

PART 3: MANDATORY ENGINEERING SAFETY FEATURES (21 CFR 1040.10(f))

Feature	21 CFR Clause	Status	Engineering Implementation Description
Protective Housing	1040.10(f)(1)	**COMPLIANT**	Enclosed CNC chassis with optical filtration hood reducing accessible radiation outside working area.
Safety Interlocks	1040.10(f)(2)	COMPLIANT	Magnetic safety interlock switches wired to microcontroller cut off diode drive current within 15 ms upon enclosure lift.
Remote Interlock Connector	1040.10(f)(3)	COMPLIANT	Standard 2-pole phoenix connector for external emergency stop button or room perimeter switches.
Key Control	1040.10(f)(4)	COMPLIANT	Physical rotary cylinder key control located on chassis front fascia; key captive in ENABLE state.
Beam Attenuator / Shutter	1040.10(f)(6)	COMPLIANT	Spring-loaded solenoid mechanical shutter closes beam aperture upon power fault or command.
Emission Indicator	1040.10(f)(5)	COMPLIANT	Dual optical amber indicators and audible chime providing 5-second advance warning before laser firing.
Viewing Optics / Window	1040.10(f)(8)	COMPLIANT	Enclosure observation pane rated OD 6+ at 400-500nm laser wavelength.

PART 4: REQUIRED STATUTORY LABELING & MARKINGS (21 CFR 1010.2, 1010.3, 1040.10(g))

Certification Label (21 CFR 1010.2)	Complies with FDA radiation performance standards, 21 CFR 1040.10 and 1040.11 except for conformance with IEC 60825-1 Ed. 3, as described in Laser Notice No. 56, dated May 8, 2019.
Identification Label (21 CFR 1010.3)	Manufactured by [Firm Name], Address: [Address], Date: [MM/YYYY], Serial No: [Serial]
Warning Logotype Label (1040.10(g))	DANGER - LASER RADIATION AVOID EYE OR SKIN EXPOSURE TO DIRECT OR SCATTERED RADIATION. CLASS IV LASER PRODUCT.
Aperture Label (1040.10(g)(4))	AVOID EXPOSURE - Laser radiation is emitted from this aperture.

Interlock Defeat Label (1040.10(g)(6))	CAUTION - Laser radiation when open and interlock defeated.

PART 5: QUALITY CONTROL, IN-LINE TESTING & CALIBRATION (21 CFR 1002.30)

Incoming Verification: 100% receipt verification of laser diode batches for optical output power and forward voltage.
Production Testing: 100% finished unit burn-in testing (4 hours continuous) followed by calibrated power meter check.
Measurement Calibration: Thorlabs PM100D Optical Power Meter calibrated annually traceable to NIST standard. (NIST-traceable: Yes)

PART 6: LEGAL CERTIFICATION AND ATTESTATION

I certify that all information contained herein is true and correct to the best of my knowledge and belief, and that the product(s) described herein will comply with all applicable radiation safety performance standards prescribed in Title 21 of the Code of Federal Regulations, Chapter I, Subchapter J - Radiological Health.

Authorized Representative: Justin Zhang	**Date:** 2026-11-02
Title: Director of Engineering & Regulatory Compliance	**FDA Accession No.:** _____